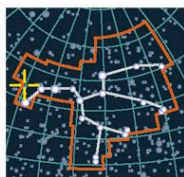


SC SPIRAL GALAXY

Pinwheel Galaxy



CATALOG NUMBERS
M101, NGC 5457

DISTANCE
27 million light-years

DIAMETER
170,000 light-years

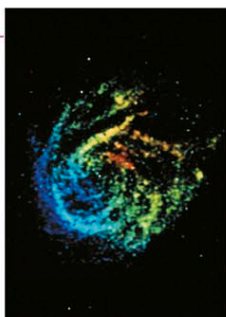
MAGNITUDE 7.9

URSA MAJOR

Cataloged by Charles Messier (see p.73) as M101, the Pinwheel Galaxy is a bright, nearby spiral galaxy, but one that reveals its nature only when studied with powerful telescopes or seen on long-exposure photographs. Because it lies face-on to Earth, most of the Pinwheel's light is spread out across its disk, and a casual glance reveals only the bright central core. Detailed photographs show that M101 has an extensive, though rather

RELATIVE RED SHIFT

This computer image shows the red shift and blue shift of objects within M101, revealing its rotation. Yellow and red regions are moving away; green and blue parts are approaching.



lopsided, spiral-arm system, giving the appearance that the core is offset from the galaxy's true center. M101 is one of the largest spirals known—its visible diameter is more than twice that of our own galaxy. Its large, angular size in the sky (larger than a full moon) makes it one of the few galaxies whose individual regions can be isolated for study.

ASYMMETRICAL DISK

M101's lopsided shape is thought to be caused by uneven distribution of mass in the disk affecting the orbit of its stars.

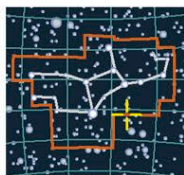


DUST LANE

The thick dust lane around the Sombrero Galaxy is silhouetted against its bright disk in this Hubble Space Telescope image.

SA SPIRAL GALAXY

Sombrero Galaxy



CATALOG NUMBERS
M104, NGC 4594

DISTANCE
50 million light-years

DIAMETER
50,000 light-years

MAGNITUDE 8.0

VIRGO

The dark dust lane and bulbous core of the Sombrero Galaxy (M104) give it a likeness to the traditional Mexican hat after which it is named. From Earth, we see the Sombrero Galaxy from 6 degrees above its equatorial plane—an ideal angle to provide a clear view of the core while also revealing the spiral arms. It is usually classified as an Sa or Sb spiral,

although its core is unusually large and bright. Another odd feature is the dense swarm of globular star clusters orbiting the galaxy. More than 2,000 have been counted—10 times more than orbit the Milky Way.

In the galaxy's core is a disk of bright material tilted relative to the galaxy's plane. It is probably the accretion disk of a central supermassive black hole. X-ray emission from the region suggests some material is still being absorbed by the hole.

M104 was a late addition to Messier's catalog of celestial objects. He added it by hand to his copy of the catalog after discovering it in 1781. Several other astronomers also found it independently. One



COMBINED VIEW

This composite image shows the Sombrero at X-ray (blue), optical (green), and infrared (orange) wavelengths.

of these was William Herschel, who was the first to note the dark dust lanes that are M104's most distinctive feature. More recently, the Sombrero provided some of the first evidence for objects lying far beyond our own galaxy (see panel, below).

VESTO SLIPHER

US astronomer Vesto Slipher (1875–1969) was one of the first to suggest that the universe is bigger than our galaxy. In 1912, at Lowell Observatory in Flagstaff, Arizona, he identified red-shifted lines in M104's spectrum. The lines told him the galaxy was receding at 2.25 million mph (3.6 million kph)—too fast for it to reside within the Milky Way.

